

Query Optimization

Based on the video, the following points are explained in detail.

1. Comparison of CHAR and VARCHAR data types

Both CHAR and VARCHAR store character strings, but they differ in how they allocate storage space.

CHAR is a fixed-length data type. When you define a column as CHAR(10) and insert the value 'SQL', the database still stores 10 characters. The remaining unused space is filled with blank spaces.

VARCHAR is a variable-length data type. When you define VARCHAR(10) and insert 'SQL', the database stores only the 3 characters entered.

2. When to use integer or numeric data types

Use numeric data types such as INT when the column stores only numbers. Example: Emp_id INT(11) NOT NULL.

Integer types improve performance and reduce storage use. They often use about 4 bytes, while character types use more space and slow down queries.

3. Differences between UNION and UNION ALL

Both UNION and UNION ALL combine results from multiple SELECT statements.

UNION sorts the result and removes duplicate rows.

UNION ALL keeps duplicates and skips sorting, which improves speed when duplicates do not matter.

4. Advantages of specific SQL features

a. DISTINCT ensures the result contains only unique rows.

b. LIMIT restricts the number of rows returned.

c. != operator filters values that are not equal.

d. <> operator performs the same function as != and follows the SQL standard across systems.

Use of the EXPLAIN Command

The EXPLAIN command acts as a diagnostic tool. It shows how the database executes a query.

It displays details such as the type column. If the value is ALL, the database performs a full table scan, which is slow.

It also shows Possible Keys and Keys to indicate index usage, Key Length for comparison speed, and Ref which ideally shows const for fastest lookup.

Actual Optimization Process

Using EXPLAIN, the issue identified was full table scans due to missing indexes in JOIN and WHERE clauses.

Indexes were added to these columns so the database no longer scans every row.

After running EXPLAIN again, the type changed from ALL to ref, showing index usage.

A unique index would perform better, but duplicates in the table prevented this.

Final Result

Execution time improved from 1 minute and 59 seconds to 1.12 seconds after optimization.

This shows the effect of proper indexing on query performance.